

MINIMAL Model

Introduction to CGE Modeling Training

Arief Anshory Yusuf

Chiang Mai University, Thailand

March 4, 2023

CES Production Function Review

MINIMAL
Model

AA Yusuf

Review CES
Production
Function

MINIMAL's
Production

MINIMAL's
Demand

Sourcing

Demand for
Composite

Zero Profit
and
Equilibrium

Closures

Database

General CES production function: $Q = \left\{ \sum_i \delta_i X_i^{-\rho} \right\}^{-\frac{1}{\rho}}$

or $Q = \left\{ \sum_i \delta_i X_i^{\frac{\sigma-1}{\sigma}} \right\}^{\frac{\sigma}{\sigma-1}}$ where $\sigma = \frac{1}{1+\rho}$ is elasticity of substitution and

$0 < \sigma < \infty$, For convenience, let's simplify this to $Q(\mathbf{X}_i; \sigma_q)$. Where $\mathbf{X}_i = X_1, X_2, \dots, X_i, \dots, X_n$

So, for example: $X = \left(\delta_K \cdot K^{\frac{\sigma_X-1}{\sigma_X}} + \delta_L \cdot L^{\frac{\sigma_X-1}{\sigma_X}} \right)^{\frac{\sigma_X}{\sigma_X-1}}$

can be written as $X(K, L; \sigma_X)$.

Remember also that:

- $X(K, L; 0)$ is Leontief production function, and
- $X(K, L; 1)$ is Cobb-Douglas production function.

CES Production Function with Technological Change

CES production fn. with technological change: $Q = \frac{1}{A_Q} \left\{ \sum_i \delta_i \left(\frac{X_i}{A_i} \right)^{-\rho} \right\}^{-\frac{1}{\rho}}$

or $Q = \frac{1}{A_Q} \left\{ \sum_i \delta_i \left(\frac{X_i}{A_i} \right)^{\frac{\sigma-1}{\sigma}} \right\}^{\frac{\sigma}{\sigma-1}}$ where $0 < \sigma = \frac{1}{1+\rho} < \infty$.

We can simplify this to $Q \left(\frac{\mathbf{X}_i}{\mathbf{A}_i}; \sigma_Q, A_Q \right)$.

So, for example: $X = \frac{1}{A_X} \left(\delta_K \cdot \frac{K}{A_K}^{\frac{\sigma_X-1}{\sigma_X}} + \delta_L \cdot \frac{L}{A_L}^{\frac{\sigma_X-1}{\sigma_X}} \right)^{\frac{\sigma_X}{\sigma_X-1}}$

can be written as $X \left(\frac{K}{A_K}, \frac{L}{A_L}; \sigma_X, A_X \right)$.

Then:

- $X \left(\frac{K}{A_K}, \frac{L}{A_L}; 0, A_X \right)$ is Leontief production function, and
- $X \left(\frac{K}{A_K}, \frac{L}{A_L}; 1, A_X \right)$ is Cobb-Douglas production function.

2x2 vs MINIMAL: Production

2x2 model production function:

$$X_i(K_i, L_i; \sigma_X)$$

With optimization and linearization:

- $k_i = x_i - \sigma_X(r - c_i)$
- $l_i = x_i - \sigma_X(w - c_i)$
- $c_i = S_i^L w + S_i^K r$

Now, define Z_{ci} intermediate input commodity c used in production of X_i :

MINIMAL model's CES *separable* production function:

$$X_i(V_i(K_i, L_i; \sigma_V, A_V), \mathbf{Z}_{ci}; 0)$$

With optimization and linearization:

- $k_i = v_i - \sigma_V(r_i - c_i)$
- $l_i = v_i - \sigma_V(w_i - c_i)$
- $c_i = S_K \cdot r_i + S_L \cdot w$
- $v_i = a_v + x_i$
- $Z_{ci} = x_i$

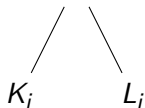
2x2 vs MINIMAL: Nested Production Function

MINIMAL
Model

AA Yusuf

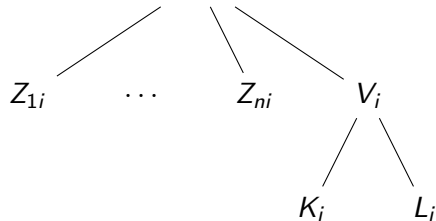
2x2 Model:

$$X_i(K_i, L_i; \sigma_X)$$



MINIMAL model:

$$X_i(V_i(K_i, L_i; \sigma_V, A_V), \mathbf{Z}_{ci}; 0)$$



MINIMAL Production Function Tablo Equations

MINIMAL
Model

AA Yusuf

Review CES
Production
Function

MINIMAL's
Production

MINIMAL's
Demand

Sourcing

Demand for
Composite

Zero Profit
and
Equilibrium

Closures

Database

Equation E_x1lab

$$(all,i,IND) \ x1lab(i) = x1prim(i) - SIGMA1PRIM(i)*[p1lab-p1prim(i)];$$

Equation E_x1cap

$$(all,i,IND) \ x1cap(i) = x1prim(i) - SIGMA1PRIM(i)*[p1cap(i)-p1prim(i)];$$

Equation E_p1prim

$$(all,i,IND) \ V1PRIM(i)*p1prim(i) \\ = FACTOR("Labour",i)*p1lab + FACTOR("Capital",i)*p1cap(i);$$

Equation E_x1 # demand for commodity composites #

$$(all,c,COM)(all,i,IND) \ x_s(c,i)= x1tot(i);$$

Equation E_x1prim # demand for primary-factor composites #

$$(all,i,IND) \ x1prim(i) = a1prim(i) + x1tot(i);$$

MINIMAL Demand for Commodities

MINIMAL
Model

AA Yusuf

Review CES
Production
Function

MINIMAL's
Production

MINIMAL's
Demand

Sourcing

Demand for
Composite

Zero Profit
and
Equilibrium

Closures

Database

MINIMAL distinguishes commodities by its origin (local Q_c^D or import Q_c^M) and demand come from more than one entity. Total demand for local commodities: Demand of intermediate goods by industry Z_{ci}^D , demand by household H_c^D , demand for capital creation (investment) I_c^D , demand by government G_c^D and demand for export E_c .

$$Q_c^D = \sum_i Z_{ci}^D + H_c^D + I_c^D + G_c^D + E_c$$

Demand for imported commodities:

$$Q_c^M = \sum_i Z_{ci}^M + H_c^M + I_c^M + G_c^M$$

Equation for Aggregation of Total Demand

Tablo equations for the following:

$$Q_c^D = \sum_i Z_{ci}^D + H_c^D + I_c^D + G_c^D + E_c$$

$$Q_c^M = \sum_i Z_{ci}^M + H_c^M + I_c^M + G_c^M$$

is:

Equation E_x0

```
(all,c,COM)(all,s,SRC) SALES(c,s)*x0(c,s)=  
sum{u,USER,USE(c,s,u)*x(c,s,u)};
```


Domestic-Import Sourcing

Entity that buys both domestic and imported goods behave by minimizing their combined (local and import) expenditure subject to CES combination of domestic and imported goods (composite of differentiated goods, non-infinity elasticity of substitution), or:

$$\text{minimize } P^D Q^D + P^M Q^M \text{ subject to } Q(Q^D, Q^M; \sigma_{DM})$$

The (linearized) solution:

$$q^D = q - \sigma_{DM} (p^D - p), \quad q^M = q - \sigma_{DM} (p^M - p) \quad \text{and} \quad p = S^D p^D + S^M p^M$$

σ_{DM} is called **Armington elasticities**.

Domestic-Import Sourcing Equations

MINIMAL
Model

AA Yusuf

Review CES
Production
Function

MINIMAL's
Production

MINIMAL's
Demand

Sourcing

Demand for
Composite

Zero Profit
and
Equilibrium

Closures

Database

$$q^D = q - \sigma_{DM} (p^D - p), \quad q^M = q - \sigma_{DM} (p^M - p) \quad \text{and} \quad p = S^D p^D + S^M p^M$$

Equation E_x

```
(all,c,COM)(all,s,SRC)(all,u,IMPUSER)
  x(c,s,u) = x_s(c,u) - SIGMA(c)*[p(c,s) - p_s(c,u)];
```

Equation E_p_s

```
(all,c,COM)(all,u,IMPUSER) p_s(c,u) =
  sum{s, SRC, SRCshr(c,s,u)*p(c,s)};
```

Demand for Composite Commodities and Exports

MINIMAL
Model

AA Yusuf

Review CES
Production
Function

MINIMAL's
Production

MINIMAL's
Demand

Sourcing

Demand for
Composite

Zero Profit
and
Equilibrium

Closures

Database

- Demand of intermediate from industries is Leontief: $z_{ci} = x_i$
- Demand of commodities by households is Cobb-Douglas:

$$h_c = y - p_c$$

- Demand for investment and government is **exogenous** or $i_c = 0$ and $g_c = 0$
- Demand for exports:

$$e_c = f_c^E - \epsilon_c \left((p_c^D - \pi) - p_c^W \right)$$

where f_c^E is export shifter, ϵ_c is export elasticity, π is exchange rate, and p_c^W is international price.

Equations for Demand for Composite Goods

In Tablo file:

Equation E_x1 # demand for commodity composites #

`(all,c,COM)(all,i,IND) x_s(c,i)= x1tot(i);`

Equation E_x3

`(all,c,COM) x_s(c,"Households") + p_s(c,"Households") = w3tot;`

Equation E_x4a `(all,c,COM) x(c,"dom","Exports") =`

`f4q(c) - ABS[EXP_ELAST(c)]*[{p(c,"dom")-phi}- pworld(c)];`

In CMF/Simulation file (Closure):

Exogenous `x_s(COM,"Investment") ;`

Exogenous `x_s(COM,"Government") ;`

Zero Profit and Market Clearing Conditions

Zero profit conditions for industries:

$$P_i^P X_i = \sum_c P_c^D Z_{ci}^D + \sum_c P_c^M Z_{ci}^M + WL_i + R_i K_i$$

Market clearing conditions for domestic commodities:

$$Q_c^D = X_c$$

Tablo equations:

```
Equation E_p1tot # cost of production = cost of all inputs #
(all,i,IND)          V1TOT(i)*[p1tot(i)+ x1tot(i)] =
    sum{c,COM,sum{s,SRC, USE(c,s,i)*[p(c,s)  + x(c,s,i)]}}
    + FACTOR("Labour",i)*[p1lab  + x1lab(i)]
    + FACTOR("Capital",i)*[p1cap(i)+ x1cap(i)];
```

```
Equation E_x1tot (all,c,COM) x1tot(c) = x0(c,"dom");
```

Producer's Price, Purchaser's Price and Taxes

Define T_c^D and T_c^M are advalorem sales tax rate that link producer's price to purchaser's price for domestic and imported commodities respectively:

$$\text{Domestic commodities} \Rightarrow P_c^D = \left(1 + \frac{T_c^D}{100}\right) P_c^P$$

$$\text{Imported commodities} \Rightarrow P_c^M = \left(1 + \frac{T_c^M}{100}\right) P_c^W$$

Tablo equations:

Equation E_pA

(all,c,COM)

$$p(c, "dom") = p1tot(c) + 100 * [V1TOT(c) / (V1TOT(c) + V1PTX(c))] * Delptxrate(c);$$

Equation E_pB

(all,c,COM)

$$p(c, "imp") = pworld(c) + phi + 100 * [V0CIF(c) / SALES(c, "imp")] * Delmtxrate(c);$$



Closures

Short-run:

```
phi           ! 1 Exchange rate $A/$US
x_s(COM,"Investment") ! COM Investment demands
x_s(COM,"Government") ! COM Government demands
x1cap         ! IND Current capital stocks
realwage      ! 1 Real wage
x3tot         ! 1 Real household consumption
a1prim        ! IND Factor-using technical change
pworld        ! COM World prices $US
f4q           ! COM Export demand shift
Delmtxrate    ! COM Tax rate on imports
Delptxrate    ! COM Tax rate on production
```

Long-run:

```
swap x1cap = gret;      ! Capital is mobile across sectors
swap realwage = employ; ! Aggregate employment is fixed
swap x3tot = delB;      ! Trade balance (per GDP) is fixed
```

Database structure of MINIMAL Model

MINIMAL
Model

AA Yusuf

Review CES
Production
Function

MINIMAL's
Production

MINIMAL's
Demand

Sourcing

Demand for
Composite

Zero Profit
and
Equilibrium

Closures

Database

		Absorption Matrix					
		1	2	3	4	5	
		Producers	Investors	Household	Export	Government	Total Sales
	Size	← I →	← 1 →	← 1 →	← 1 →	← 1 →	
Domestic Flows	↑ C ↓	USE(commodity,"dom",user)					
Imported Flows	↑ C ↓	USE(commodity,"imp",user)					
Labour	↑ 1 ↓	FACTOR (labour)	C = Number of Commodities = 7 I = Number of Industries = 7				
Capital	↑ 1 ↓	FACTOR (capital)					
Output tax	↑ 1 ↓	V1PTX					

		Tax on Imports
Size	← 1 →	
↑ C ↓	V0MTX	